VIGENTRA

VIGILANCE | INTELLIGENCE | SAFER ROADS

GUJARAT POLICE INNOVATION CHALLENGE 2026 · CCTV INTEGRATION HACKATHON

One platform over Gujarat's CCTV — without asking 26 departments to replace theirs

A federated camera registry and GIS, live analytics at the edge, watchlist alerts and cross-camera vehicle tracking, running today on the 30-camera government feed.

Solution presentation

September 2026

THE PROBLEM

Many systems, no shared view, no shared memory

26

government departments, each with its own CCTV and VMS

~80,000

cameras to reach statewide, analog and IP, many vendors

~1,000 km

between the furthest sites, on links that are not uniform

- Vendors, AMC periods and storage differ; retention runs from 7 days to 15 or more.
- No department can see another's cameras, or check what a camera sees against the records that would make it actionable.
- The brief: one searchable platform, analysed and alerting in real time, that grows toward 80,000 cameras without a redesign.

The constraint that shapes everything

Departments keep running their own systems. The first department to join gets value on day one, and the twenty-sixth changes nothing for the first.

MODEL CHOSEN, AND WHY

A hybrid: Model 1 foundation, Model 3 integration, Model 2 console

1

Registry & GIS

Adopted in full

Mandatory, and correct: nothing else works without knowing which cameras exist, where, and who owns them. Useful on day one with no analytics at all.

3

VMS federation

The integration

One adapter per vendor, normalising to one camera model. Vendor differences stop at the adapter; a department joins without touching anything downstream.

2

Unified viewing

As the surface

One console for operators — but it talks to the platform, never directly to 26 VMS, so a new vendor is never a change to the screen people use.

4

Central VMS

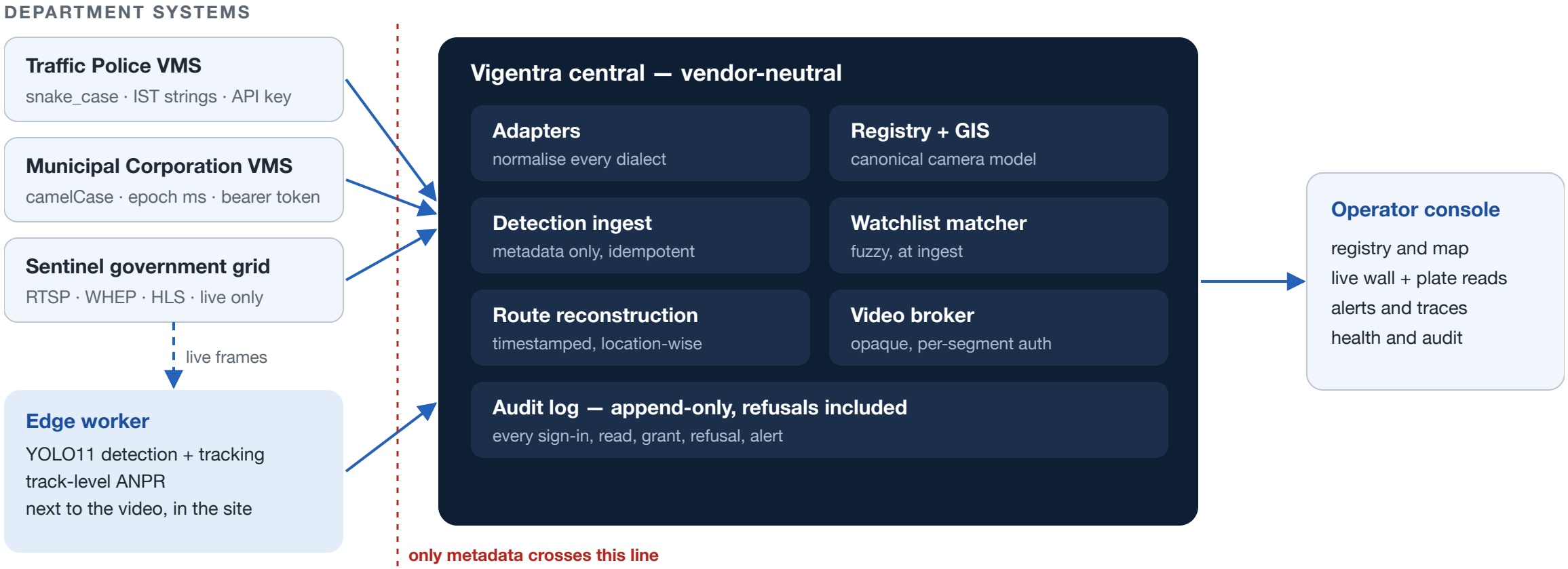
Rejected

Streaming 80,000 cameras centrally is ~160 Gbps and ~1.7 PB a day, and puts every camera credential in one place. Inference goes to the video instead.

Video is pulled on demand, per authorised session. Analytics run next to the cameras and send only metadata.

SOLUTION OVERVIEW

Frames stay where they are; metadata travels



INTEGRATING HETEROGENEOUS CCTV

One adapter per vendor; above it, every camera looks the same

	TRAFFIC POLICE	MUNICIPAL CORPORATION	SENTINEL GRID
Fields	flat snake_case	nested camelCase	minimal catalogue
Time	IST wall-clock strings	epoch milliseconds	absent — PTS from the stream
Auth	API key header	bearer token	one session per account
Register / archive	yes / yes	yes / yes	none / none — read-only, live-only in code

Analog and legacy

An analog camera is reached through the DVR or encoder that digitises it, which exposes RTSP or ONVIF like any IP source.

Outages stay local

One department down leaves the rest working; its cameras go offline with a stated reason, and every failed poll is audited.

Liveness follows the video

If the grid's web gateway refuses but its video server answers, cameras stay online — this recovered 72 of 180 lost minutes on 14 Sep.

VIDEO ANALYTICS

ANPR that reads a vehicle, not a frame

1 · Detect & track
YOLO11 + ByteTrack give each vehicle a stable id

›

2 · Find the plate
only on vehicles ≥ 96 px wide, where a plate can be read

›

3 · Bank the crops
every crop stamped with frame, PTS and quality

›

4 · Restore
rectify, register, fuse, deglare, deblur

›

5 · Read
CRNN readers on every variant; a second reader fills gaps

›

6 · Vote
Indian plate grammar + consensus across the track

Grammar, not guesswork

A character's position says letter or digit, so **GJ05JV34S6** can only be GJ05JV3456. Impossible states and districts are dropped at the edge.

One vehicle, one reading

Emitted once per vehicle with the number of frames that agreed. Unconfirmed readings are shown as unconfirmed, never as fact.

Object detection

Person, car, motorcycle, bus, truck, auto-rickshaw, bicycle — with a frame-quality router that skips frames too degraded to mean anything.

Skipping plate search on vehicles too small to read gave the same 36 readings on the labelled Delhi clip in 586 s instead of 878 s.

WATCHLIST ALERTS AND VEHICLE TRACKING

Known at the moment of the read; traced across cameras

!

Matched at ingest

- Every plate reading is checked against the active watchlist as it arrives — stolen, wanted, missing, blacklisted.
- Fuzzy on purpose: a known OCR confusion (B↔8, O↔0) costs 0.35, an unrelated character 1.0.
- Alerts say whether the match was exact, and near matches are shown, not hidden.

→

The route, from the plate

- Given a registration: every permitted sighting in time order, with distance, elapsed time and speed per leg, on the map.
- Repeated reads at one camera collapse into one pass; impossible legs are flagged, never dropped.
- Every trace needs a written reason, recorded against the account.

Seconds, not minutes

A plate is uploaded with the frame that settled it; the console refreshes every 5 s.

Prioritised

critical exact stolen or wanted · **high** any other exact · **review** near match. Open alerts lead, most urgent first.

Leaves the screen

Each alert is also POSTed to a webhook, HMAC-signed — dispatch, SMS or messaging sit behind it.

Closed by a person

Acknowledged or dismissed with a reason; raise, acknowledge and dismiss are all audited.

The test case is a designated vehicle's complete, timestamped route. Cameras whose plates are legible get a continuous reader: a sampled camera does not see a vehicle that crosses in five seconds.

CYBERSECURITY, PRIVACY AND AUDIT

Nobody sees more as cameras are added

1

Brokered, opaque video

Short-lived, watermarked sessions; every segment is re-authorised, so revoking a grant stops a playing stream.

2

Scope in three dimensions

Department, city and zone, checked per request, plus each camera's own access policy.

3

Least privilege

13 roles, 22 permissions. The account that ingests plates cannot read the watchlist back.

4

Withheld, not refused

Without plate permission an officer still gets the alert, with the plate withheld.

5

Retention on read

Plates keep a shorter clock, applied at read time — a failed purge cannot extend it.

6

A complete audit trail

Every sign-in, read, grant, refusal and machine-raised alert, each committed on its own.

WORKING ON THE GOVERNMENT FEED — MEASURED 14–20 SEPTEMBER 2026

Live, measured, and honest about the limits

30/30
grid cameras onboarded from the catalogue, online

36/48
plates a person can read, read correctly, across 38 recorded clips (was 9 of 26)

0
false plates confirmed or sent, on any clip, day or night

332
automated tests passing (201 platform, 131 edge)

Plates read per 2 minutes, grid cam06, by the recording's clock



At noon, 15 of the 16 plates legible by eye were read and none was wrong; the sixteenth is misread by every reader and is refused.

Ground truth is by eye and used only for scoring. At night these 720p feeds carry plates 14–31 px wide: nobody can read them, and the system reads none rather than guessing. A false plate costs more than a missed one — that is why an alert from this system can be acted on.

THE SENTINEL INTEGRATOR'S GUIDE, ITEM BY ITEM

Built against the grid as it behaves, not as a file

RTSP over TCP

forced in every client; HLS only when 8554 is blocked

Timing from PTS

never arrival time; the declared frame rate is never trusted, the delivered rate is measured

Reconnect with backoff

2 s to 30 s; a failing camera keeps the full pause, never a tight loop

Decoder warnings on join

logged, not fatal; mixed H.264 and H.265

The catalogue is the contract

cameras and their properties come from `/api/ingest`; nothing is hard-coded

No uniform grid

each camera detected at its own resolution, with its own profile in configuration

Scene discontinuity

tracks and motion state reset at the loop cut; learned traffic flow survives

Consume only, paced

a bounded pool of streams; nothing published, no control API

Mapped to the code that does each one in `docs/sentinel-grid.md`.

WHAT EACH MODULE DOES · 1 OF 2 · AT THE CAMERA

The edge reads the video so nothing else has to

Live reader

Reads every grid camera at once on one GPU; cameras with a vehicle near the lens get it first. A supervisor restarts it after a crash, a grid outage or idle cameras, and starts it at boot.

ANPR engine

Finds each vehicle's plate, keeps its best crops, enhances them, reads them with two readers and an Indian-plate grammar, and votes across frames. Unsure reads are kept as candidates, never as facts.

Vehicle types, Indian classes

Scooter, motorcycle, auto, car/jeep (M1), minibus (M2), bus (M3), light goods (N1: Tata Ace, pickup), truck. Drops sign boards and barricades the detector takes for buses. 86 % right against 50 %.

Offence analytics

Rider without a helmet (4 classifiers, one call per rider, replays remembered) and wrong way; collision, person in traffic and intrusion as candidates. Every call carries its snapshot.

Pursuit check

While a vehicle is followed, every vehicle at the cameras it can reach is tested against its plate, so a half-read plate can still raise a "possible" for a person to confirm.

Footage quality

Flags frames that are corrupted, black, dark, glared or blurred; a broken picture raises a camera alert within a minute. Only text, boxes and crops leave the site, never video.

Central holds the facts; the console puts a person in charge

Federation adapters

One adapter per department system or vendor turns its API into one camera model; departments keep their own VMS, footage and retention.

Registry, GIS and gap analysis

Every camera on one map with owner, purpose and health; districts that are thin, cameras that are ageing, and footage that is online but unusable.

Watchlist and alerts

Each plate read is matched at ingest, allowing for OCR confusions (0/O, 8/B); an alert is raised in seconds, prioritised, and closed by a person.

Trace and pursuit

A plate and a reason give the route on the map: camera, place, time read and time on the footage, how many frames agreed; impossible legs flagged.

Incident review and e-challan

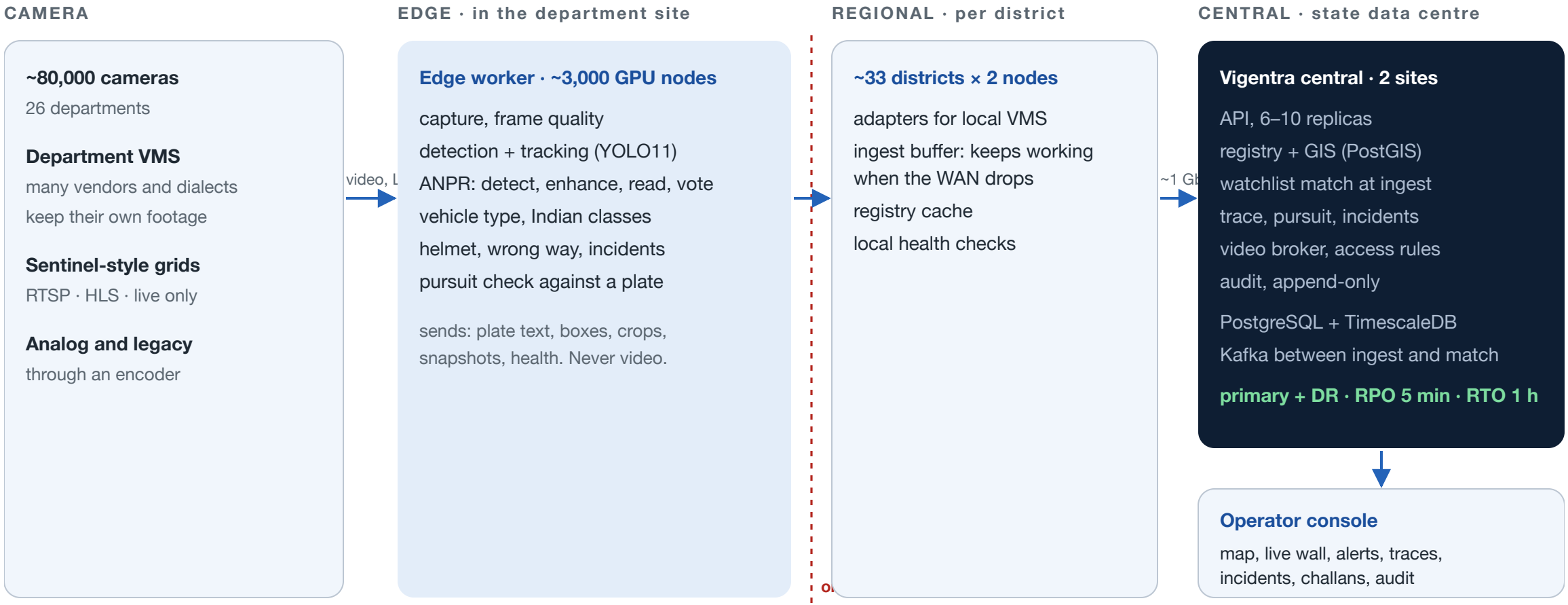
Candidates as cards led by the snapshot; Confirm opens the challan with the read plate filled in, owner looked up, SMS sent (demo stand-ins for VAHAN and the gateway).

Video broker, access and audit

Live video only through an audited session, by role, department and zone; grants between units; every sign-in, trace, lookup and refusal in an append-only log.

HIGH-LEVEL DESIGN

Four tiers; only metadata crosses the line



AT 80,000 CAMERAS · WHAT BECOMES POSSIBLE

One query instead of 26 footage requests

Statewide vehicle trace

A named plate's route across every department's cameras, in minutes, with the evidence crop at each point.

Live pursuit across districts

The next cameras a vehicle can reach get priority as it moves; route corridors on PostGIS predict where to look.

Watchlist at every junction

Stolen and wanted vehicles flagged at the moment of the read, across the whole state; a bloom filter keeps millions of entries cheap.

Automated e-challan

Helmet, wrong side, red light, triple riding, no seat belt: candidates with evidence, a person confirms, VAHAN gives the owner, SMS goes out.

Traffic planning by vehicle class

Counts and origin-destination flows of cars, autos, buses, light and heavy goods vehicles per junction and hour; goods vehicles in no-entry hours.

A camera estate that maintains itself

Every camera's health and footage quality watched; outages, black screens and broken pictures raised to the owning department the same minute.

Built and running today on the 30-camera grid: trace, pursuit, watchlist, helmet and wrong way with the challan flow (demo owners), Indian vehicle classes, footage alerts. The rest extends the same pipeline.

AT 80,000 CAMERAS · WHAT IT NEEDS

The requirements, tier by tier

NEED	REQUIREMENT (ESTIMATE, FOR PLANNING)
Edge compute	~3,000 GPU nodes in department sites (1× T4-class GPU, 8 vCPU, 32 GB RAM each): object detection everywhere, ANPR only where the optics allow, classified at onboarding (~15 ANPR cameras or ~30 detection cameras per GPU)
Regional	33 districts × 2 nodes (16 vCPU, 64 GB RAM, 2 TB NVMe): adapters, ingest buffer for WAN outages, registry cache
Central	2 sites, primary + DR: API 6–10 replicas; PostgreSQL 2 × 32 vCPU / 256 GB / 10 TB NVMe with read replicas; object storage for backups and cold data
Network	~0.6–1 Gbps statewide into the data centre (metadata only; central video would need ~160 Gbps); camera-to-edge video stays on the site LAN
Storage	No video. Metadata hot 30 days, warm 1 year, cold 7 years; plates on a shorter clock (30 days by default); audit log kept in full
Integrations	Each department's VMS API and credentials; VAHAN owner lookup (MV Act s.136A, CMVR r.167A); a DLT-registered SMS gateway; the state network (GSWAN) to every site
People	District operators to confirm alerts and incidents; a monitoring desk for camera health; a small team labelling footage to retrain the models
Cost	Edge accelerators ₹90–120 crore over the rollout phases; all software open source, no licences. Avoided: 160 Gbps of WAN, ~1.7 PB of video a day, replacing 26 VMS

PLAN FOR SCALE

To ~80,000 cameras without moving the video

Streaming everything centrally

160 Gbps

sustained, and ~1.7 PB of video per day

Vigentra: metadata only

0.6–1 Gbps

statewide, 40–250× less; departments keep their footage

~3,000 GPUs

ANPR only where the optics allow, classified at onboarding (estimate)

Hot · warm · cold

30 days, 1 year, 7 years; plates on a shorter clock

RPO 5 min · RTO 1 h

synchronous standby, point-in-time recovery, DR site

Watched at scale

per-camera health history, liveness from the video, audited

Named and planned: trigram prefilter for fuzzy plate search, bloom filter for the watchlist, PostGIS for corridor queries.

STATEWIDE ROLLOUT

Each phase is useful on its own

Phase 1 · ~50 cameras
one district, all four models exercised — this hackathon

›

Phase 2 · ~2,000
3 districts, 2 departments — genuinely different VMS vendors

›

Phase 3 · ~15,000
all districts, Home Department — statewide load, DR failover tested

›

Phase 4 · ~80,000
all 26 departments — steady state

Model 1 goes first, in full, at every phase

An accurate statewide camera inventory with coverage-gap analysis is valuable on the day it exists — and nobody has one today. Analytics follow the inventory.

BEYOND THE MANDATORY

Built, running, and verifiable in the code

Hybrid architecture
Models 1, 3 and 2 together, each where it fits

Cross-camera tracking
routes from plate reads, impossible legs flagged

Analytics beyond ANPR
intrusion zones, wrong-way, stopped in lane, person in traffic

Edge processing
inference beside the video, on CPU or GPU

Bandwidth optimisation
metadata only; video on demand; sampling dial

Security and audit
brokered sessions, fine roles, append-only trail

Automated alerts
matched at ingest, prioritised, signed webhook out

Health monitoring
per-source and per-camera, with history

Operational dashboards
registry, map, live wall, alerts, reports

Integration-ready APIs
documented REST API and adapter contract

OUR PERSPECTIVE

What we believe — and what the grid taught us

Start from the people who use it

A control-room officer needs one screen; twenty-six departments should not have to change theirs to give it to them. Federation, not replacement, is what makes that possible on day one.

Never show more certainty than we have

An unconfirmed reading is shown as unconfirmed, a plate too small to read is marked unreadable instead of guessed, and every figure in this deck was measured or is labelled as an estimate.

Privacy is the design, not a feature

Footage is asked for, never ambient. Every look is brokered and audited — refusals included — because the thing that scales worst in surveillance is how many people can see everything.

The hard part was the plumbing

On the government feed the model was rarely the problem: refused sessions, dropped streams and a shared account were. The platform now keeps cameras online while video flows and restarts its own readers.

VIGENTRA

VIGILANCE | INTELLIGENCE | SAFER ROADS

This is what we were able to build in the time we had.

With upgraded hardware, the same software can do much better.

Today

One laptop — an Apple M1 with 16 GB. Our readers share its single GPU and visit the 30 cameras in turn, a few minutes each.

With edge GPUs

About one T4-class card per 15 cameras (estimate): every camera read continuously, heavier models, more frames of every vehicle.

With cameras placed for ANPR

Plates large enough to confirm rather than guess — on today's wide junction views it is the optics, not the code, that set the ceiling.

Source: github.com/rewardix-ai/vigentra · Design: docs/hld.md · Scale: docs/scalability.md · Demonstration videos and output report: links in the submission